

Individual Assignment 5

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Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c(" ", "n/a", "N/A", "over", "173+")
)

```

Cleaning Data

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

```

```

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames

```

```

unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

```

```

aq_ins_clean <- aq_ins |>
clean_names() |>
semi_join(ncos_sites, by = "site") |>
rename (date = date_on_vial) |>
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
filter(sample_type %in% c("FB250", "FB 250")) |>
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>
group_by(date, site, taxon) |>
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list

```

```

left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
))

```

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median salinity (ppt)",
  title = "Site salinity") +
# different theme
theme_bw()

```

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

```

```

# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

```

1. Visualizing differences across sites in major taxon abundance

a)

- x-axis: site
- y-axis: log + 1 transformed average abundance/L
- geom to represent average abundance/L: `geom_jitter()`
- geom to represent medians: `stat_summary()`

b)

```

# creating new object from clean aquatic invertebrates
ostracod <- aq_ins_clean |>
# filter to only include ostracod
filter(taxon == "ostracod") |>

```

```
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

ostracod |>
  slice_sample(n = 10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2020-01-21_NDC	2020-01-21 00:00:00	NDC	ostr~	2.53	6.53	0.3	7.6
2	2022-11-19_NPB1	2022-11-19 00:00:00	NPB1	ostr~	2.13	7.26	2.47	3.96
3	2023-02-28_NPB	2023-02-28 00:00:00	NPB	ostr~	2.67	NA	NA	NA
4	2022-01-21_NEC	2022-01-21 00:00:00	NEC	ostr~	NA	NA	NA	NA
5	2023-11-14_NPB	2023-11-14 00:00:00	NPB	ostr~	18.1	NA	NA	NA
6	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.27	NA	NA	NA
7	2021-11-12_NPB	2021-11-12 00:00:00	NPB	ostr~	79.3	NA	NA	NA
8	2022-08-08_NPB2	2022-08-08 00:00:00	NPB2	ostr~	2.13	7.68	2.36	2.43
9	2022-01-24_NEC	2022-01-24 00:00:00	NEC	ostr~	0.4	NA	NA	NA
10	2022-08-12_NVBR	2022-08-12 00:00:00	NVBR	ostr~	NA	NA	NA	NA

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
comments <chr>, site_full <chr>, log_ave_lit <dbl>

c)

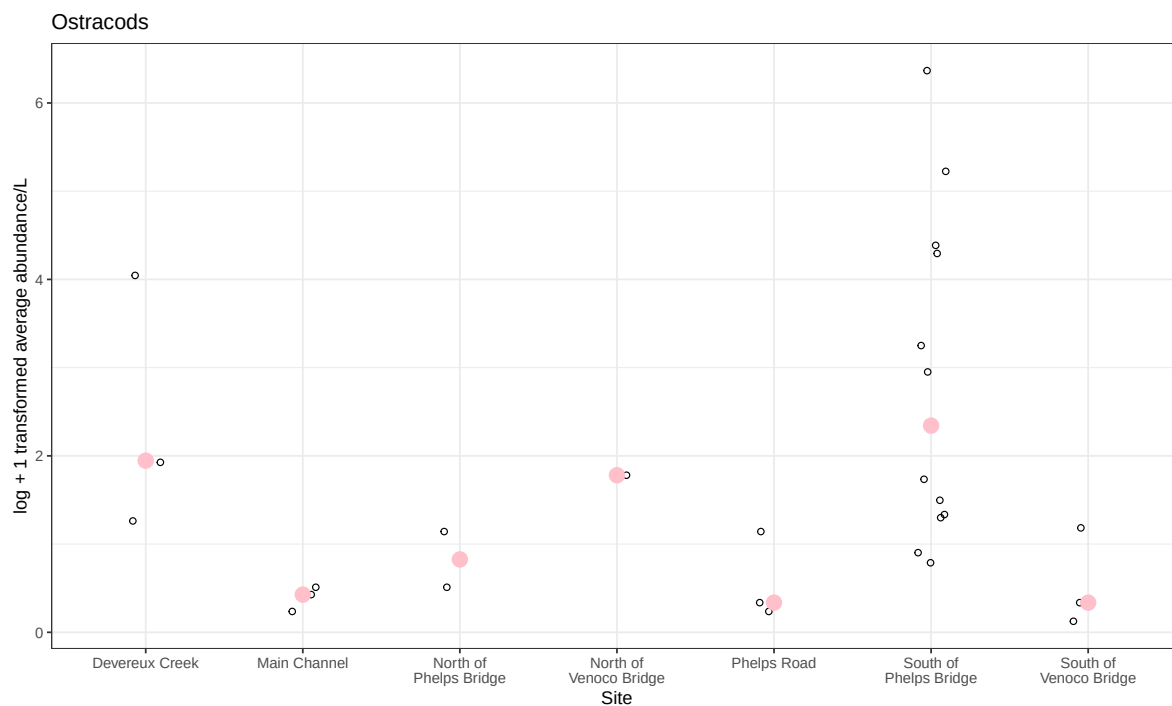
```
# base layer
ostracod_plot <- ggplot(data = ostracod,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
```

```

        color = "pink") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
      y = "log + 1 transformed average abundance/L",
      title = "Ostracods") +
# different theme
theme_bw()

# displaying object
ostracod_plot

```

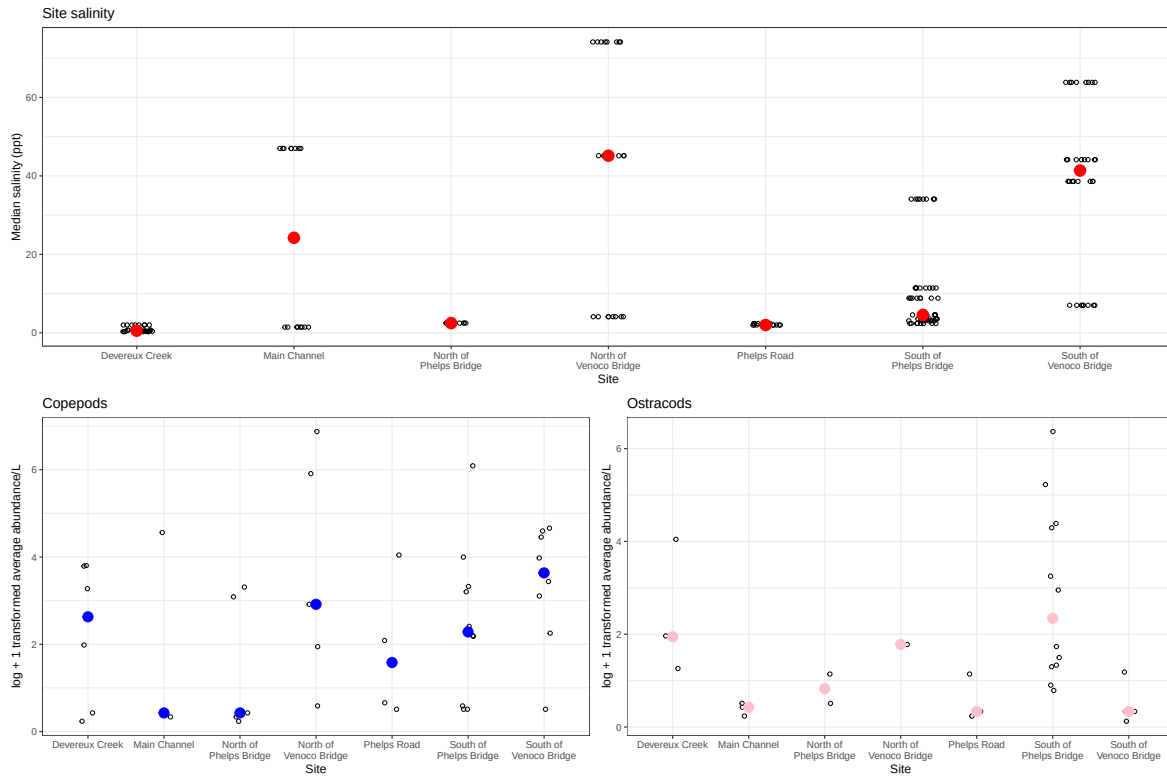


d)

```

# create multipanel figure
salinity / (copepods + ostracod_plot)

```



e)

Copepods are more abundant than ostracods at these sites: Devreux Creek, North of Venoco Bridge, Phelps Road, and South of Venoco Bridge.

It does not appear that ostracod abundance relates to site-level differences in salinity. At South of Venoco Bridge salinity is high while abundance is low however at North of Venoco Bridge both salinity and abundance are high.